

Newton's Laws

ANSWER KEY



Newton's first law and equilibrium

- 1 A book rests motionless on a table. State Newton's first law, and identify the two forces acting on the book that must be balanced.

Newton's first law states that an object remains at rest, or moves with constant velocity, unless acted on by a resultant (unbalanced) force. The two forces on the book are its weight (acting downward) and the normal reaction from the table (acting upward), which are equal in magnitude and opposite in direction.
- 2 A ball hangs motionless from a single string attached to the ceiling. Identify the two forces acting on the ball and explain why they must be equal and opposite.

The two forces are the ball's weight (acting downward) and the tension in the string (acting upward). Since the ball is in equilibrium (not accelerating), Newton's first law requires these forces to be equal in magnitude and opposite in direction.
- 3 A crate is pushed at constant velocity across a rough floor. Explain, using Newton's first law, why the applied force must equal the friction force.

Since the crate moves at constant velocity, its acceleration is zero, so by Newton's first law the resultant force on it must be zero. This means the applied force pushing the crate must exactly balance (equal in magnitude, opposite in direction to) the friction force resisting its motion.

Newton's second law: $F = ma$

- 4 A resultant force of 30 N acts on an object of mass 4 kg. Find its acceleration.

$$a = \frac{F}{m} = \frac{30}{4} = 7.5 \text{ m/s}^2.$$
- 5 A car of mass 1200 kg accelerates from rest to 24 m/s in 8 seconds. Find the resultant force required.

$$a = \frac{v - u}{t} = \frac{24 - 0}{8} = 3 \text{ m/s}^2. F = ma = 1200(3) = 3600 \text{ N}.$$
- 6 An object of mass 6 kg accelerates at 4.5 m/s^2 . Find the resultant force acting on it.

$$F = ma = 6(4.5) = 27 \text{ N}.$$

Weight and mass

- 7 An object has a mass of 12 kg. Find its weight on Earth, using $g = 9.8 \text{ m/s}^2$.

$$\text{Weight} = mg = 12(9.8) = 117.6 \text{ N}.$$
- 8 An object weighs 88.2 N on Earth. Find its mass, using $g = 9.8 \text{ m/s}^2$.

$$m = \frac{W}{g} = \frac{88.2}{9.8} = 9 \text{ kg}.$$

- 9 An object has a mass of 25 kg. Find its weight on Earth, using $g = 9.8 \text{ m/s}^2$.
Weight $= mg = 25(9.8) = 245 \text{ N}$.
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Multiple forces and the resultant force equation

- 10 A sled of mass 15 kg is pulled across ice by a force of 60 N, while friction of 12 N opposes the motion. Find the sled's acceleration.
Resultant force $= 60 - 12 = 48 \text{ N}$. $a = \frac{F}{m} = \frac{48}{15} = 3.2 \text{ m/s}^2$.
- 11 A crate of mass 25 kg is pushed across a floor by a force of 90 N, while friction of 18 N opposes the motion. Find the crate's acceleration.
Resultant force $= 90 - 18 = 72 \text{ N}$. $a = \frac{F}{m} = \frac{72}{25} = 2.88 \text{ m/s}^2$.
- 12 A box of mass 10 kg is pushed across a floor by a force of 45 N, while friction of 5 N opposes the motion. Find the box's acceleration.
Resultant force $= 45 - 5 = 40 \text{ N}$. $a = \frac{F}{m} = \frac{40}{10} = 4 \text{ m/s}^2$.
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Newton's third law

- 13 A swimmer pushes backward against the water with a force of 200 N. State Newton's third law, and describe the reaction force in this situation.
Newton's third law states that for every action force, there is an equal and opposite reaction force. Here, the water pushes forward on the swimmer with a force of 200 N, propelling them through the water.
- 14 A rocket engine expels exhaust gas downward with a force of 50000 N. State Newton's third law, and describe the reaction force propelling the rocket.
Newton's third law states that for every action force, there is an equal and opposite reaction force. Here, the engine pushes the exhaust gas downward with a force of 50000 N (the action), so the gas pushes back on the rocket with an equal and opposite reaction force of 50000 N, propelling it upward.
- 15 A person stands on the ground. Identify the action-reaction pair of forces between the person and the Earth, as described by Newton's third law.
The person exerts a downward force on the Earth (their weight, due to gravity), and by Newton's third law, the Earth exerts an equal and opposite upward force on the person. These two forces form the action-reaction pair — they act on different objects (the person and the Earth) and never cancel each other out.
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Motion in a lift (elevator problems)

- 16** A person of mass 70 kg stands in a lift that accelerates upward at 2 m/s^2 . Find the normal reaction force (apparent weight) from the lift floor, using $g = 9.8 \text{ m/s}^2$.
Using $F = ma$ upward: $R - mg = ma$, so $R = m(g + a) = 70(9.8 + 2) = 70(11.8) = 826 \text{ N}$.
- 17** The same person now stands in a lift accelerating downward at 1.5 m/s^2 . Find the new normal reaction force.
 $R - mg = -ma$ (acceleration downward), so $R = m(g - a) = 70(9.8 - 1.5) = 70(8.3) = 581 \text{ N}$.
- 18** A person of mass 60 kg stands in a lift that is moving at constant velocity. Find the normal reaction force from the lift floor, using $g = 9.8 \text{ m/s}^2$.
Since the lift moves at constant velocity, its acceleration is zero, so $R = mg = 60(9.8) = 588 \text{ N}$.

Connected particles with Newton's second law

- 19** A truck of mass 2000 kg tows a trailer of mass 500 kg using a light rigid tow bar. The engine produces a driving force of 5000 N, and resistance forces of 400 N (truck) and 100 N (trailer) act against the motion. Find the system's acceleration and the tension in the tow bar.
Total resultant force on the system: $5000 - 400 - 100 = 4500 \text{ N}$. Total mass: 2500 kg.
 $a = \frac{4500}{2500} = 1.8 \text{ m/s}^2$. For the trailer alone: $T - 100 = 500(1.8)$, so $T = 900 + 100 = 1000 \text{ N}$.
- 20** A car of mass 900 kg tows a trailer of mass 300 kg using a light rigid tow bar. The engine produces a driving force of 2650 N, and resistance forces of 150 N (car) and 100 N (trailer) act against the motion. Find the system's acceleration and the tension in the tow bar.
Total resultant force on the system: $2650 - 150 - 100 = 2400 \text{ N}$. Total mass: 1200 kg. $a = \frac{2400}{1200} = 2 \text{ m/s}^2$. For the trailer alone: $T - 100 = 300(2)$, so $T = 600 + 100 = 700 \text{ N}$.
- 21** A van of mass 1200 kg tows a trailer of mass 300 kg using a light rigid tow bar. The engine produces a driving force of 3300 N, and resistance forces of 200 N (van) and 100 N (trailer) act against the motion. Find the system's acceleration and the tension in the tow bar.
Total resultant force on the system: $3300 - 200 - 100 = 3000 \text{ N}$. Total mass: 1500 kg. $a = \frac{3000}{1500} = 2 \text{ m/s}^2$. For the trailer alone: $T - 100 = 300(2)$, so $T = 600 + 100 = 700 \text{ N}$.

A well-designed word problem: a rocket-powered test sled

- 22** This question has two parts. A test sled of mass 800 kg is launched from rest by a rocket motor that provides a constant thrust of 12000 N. Air and track resistance provide a constant opposing force of 1600 N throughout the motion. (a) Find the sled's acceleration. (b) Find the sled's velocity after it has traveled 50 meters, using $v^2 = u^2 + 2as$.
(a) Resultant force = $12000 - 1600 = 10400 \text{ N}$. $a = \frac{10400}{800} = 13 \text{ m/s}^2$. (b)
 $v^2 = 0^2 + 2(13)(50) = 1300$, so $v = \sqrt{1300} \approx 36.1 \text{ m/s}$.